

# TASK 3: AUTOMATION SYSTEM DESIGN

## Automated Conveyor Belt System with Sensors

*Flowchart, Circuit Diagram, Component Specification & Explanation Report*

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## 1. Introduction

Automated conveyor belt systems are the backbone of modern manufacturing, food processing, pharmaceutical production, and logistics operations. By integrating sensor technology with programmable logic controllers (PLCs) and actuators, these systems enable continuous, reliable, and intelligent material handling with minimal human intervention.

This report presents the complete design of an automated conveyor belt system equipped with proximity, weight, and optical sensors for object detection, quality inspection, sorting, and jam detection. The system is described through a detailed process flowchart, circuit/wiring diagram, component specifications, and a system explanation narrative.

## 2. System Overview

The designed system is a single-lane automated conveyor with four functional zones: (1) Object Loading Zone, (2) Inspection Zone, (3) Sorting/Divert Zone, and (4) Output Zone. Products travel on a 3-metre belt driven by a 3-phase AC induction motor via a variable frequency drive (VFD) for speed control.

### 2.1 System Objectives

- Detect the presence of objects at each zone using proximity sensors.
- Weigh objects using an integrated load cell assembly and reject underweight/overweight items.
- Inspect surface quality using a photoelectric barrier sensor array.
- Automatically divert defective or mismatched items to a reject chute.
- Stop the belt on jam detection, trigger an alarm, and alert the operator via HMI.

### 3. Component Specifications

| Component            | Model/Type              | Specification                  | Function                  |
|----------------------|-------------------------|--------------------------------|---------------------------|
| PLC                  | Siemens S7-1200         | CPU 1214C, 14 DI/10 DO         | Central control logic     |
| HMI Panel            | Siemens KTP700 Basic    | 7" colour touchscreen          | Operator interface        |
| VFD                  | Siemens SINAMICS G120   | 0.75 kW, 0–50 Hz               | Belt speed control        |
| AC Motor             | SEW EURODRIVE           | 0.55 kW, 1400 rpm, IE3         | Belt drive                |
| Prox. Sensor (x4)    | Balluff BES M08         | Inductive, NPN, 8 mm range     | Object presence detection |
| Load Cell            | HBM U2B                 | 0–10 kg, $\pm 0.02\%$ FS       | Weight measurement        |
| Photoelectric Sensor | Sick WL12L-2B430        | Retro-reflective, 0–5 m        | Size/profile inspection   |
| Reject Actuator      | Festo DSNU-25-80-P-A    | Pneumatic cylinder, 25 mm bore | Diverts rejects           |
| Jam Sensor           | Omron E2A-M08KS02-WP-B1 | Inductive, S/S, 2 mm range     | Jam/stall detection       |
| Indicator Stack      | Patlite LR6-3DZJBW      | Red/Yellow/Green LED tower     | Status indication         |
| E-Stop               | Pilz PNOZ x2.8P         | Safety relay, Cat 3/PLd        | Emergency stop            |
| 24 VDC PSU           | Phoenix Contact QUINT4  | 24 VDC, 10 A, DIN rail         | Control circuit power     |

### 4. System Process Flowchart

The following describes the step-by-step logic of the automated conveyor system. This flowchart would be drawn in Visio or Lucidchart and submitted as a separate diagram file. The textual representation below covers all logic branches:

#### 4.1 START Sequence

1. [START] → System Power ON → PLC Boot & Self-Test → HMI Displays 'READY'
2. Operator presses START on HMI → PLC checks E-Stop status → If E-Stop OK: enable VFD → Belt starts at 0.3 m/s (low speed ramp-up over 2 s)

#### 4.2 Main Operating Loop

3. Proximity Sensor 1 (PS1) at Load Zone detects object → PLC sets 'Object Present' flag
4. Belt runs at set speed (default 0.5 m/s) → Object reaches Weight Station

5. Load cell samples weight for 500 ms → Compare to Min/Max limits (e.g., 400–600 g)
6. [IF Weight OK] → Set 'Weight Pass' flag → Continue to Inspection Zone
7. [IF Weight FAIL] → Set 'Reject' flag → Activate reject actuator at Sorting Zone
8. PS2 at Inspection Zone detects object → Activate photoelectric array for 1 s
9. [IF Dimensions OK] → Maintain 'Pass' status → Continue to Output Zone
10. [IF Dimensions FAIL] → Override to 'Reject' flag
11. PS3 at Sorting Zone detects object → Check 'Reject' flag
12. [IF Reject flag SET] → Extend pneumatic diverter → Object diverts to reject chute → Retract diverter
13. [IF Reject flag CLEAR] → Object passes to Output Zone → PS4 confirms arrival
14. Production counters increment (Good/Reject) → Update HMI display

### 4.3 Fault Handling Branch

15. Jam Sensor monitors belt tension continuously → If belt stall detected for >1 s: STOP belt → Yellow light ON → HMI alarm 'Belt Jam'
16. E-Stop pressed at any time → Immediate belt stop → PLC safe state → Red light ON → Alarm latched
17. Fault reset by operator via HMI → PLC re-runs self-test → Belt restart only if fault cleared

## 5. Circuit and Wiring Diagram Description

The electrical system comprises three main circuits: the main power circuit (3-phase 400 VAC for the motor/VFD), the control circuit (24 VDC for PLC, sensors, HMI), and the safety circuit (E-stop, safety relay). A full wiring diagram is created in EPLAN Electric P8 or AutoCAD Electrical.

### 5.1 Main Power Circuit

- 3-phase 400 VAC supply feeds a 3-pole MCB (motor protection) → VFD input terminals L1/L2/L3.
- VFD output terminals U/V/W connect to AC motor terminals.
- Motor thermistor (PTC) wired to VFD thermistor input for thermal protection.

### 5.2 PLC I/O Wiring

#### Digital Inputs (DI) — PLC CPU 1214C:

- DI0.0: Proximity Sensor 1 (PS1) — Loading Zone
- DI0.1: Proximity Sensor 2 (PS2) — Inspection Zone
- DI0.2: Proximity Sensor 3 (PS3) — Sorting Zone
- DI0.3: Proximity Sensor 4 (PS4) — Output Zone
- DI0.4: Jam Detection Sensor
- DI0.5: E-Stop Safety Relay NC contact
- DI0.6: HMI Start Button
- DI0.7: HMI Stop Button

**Digital Outputs (DO) — PLC CPU 1214C:**

- DO0.0: VFD Enable (24 VDC)
- DO0.1: Reject Pneumatic Actuator Solenoid
- DO0.2: Green Stack Light
- DO0.3: Yellow Stack Light
- DO0.4: Red Stack Light / Alarm Buzzer

### 5.3 Analogue I/O (Load Cell)

Load cell signal conditioner (HBM AE101) outputs 4–20 mA proportional to weight. Connected to PLC analogue input module SM 1231 AI4 (0–20 mA range). PLC scales input to engineering units (grams) using linear scaling block in TIA Portal.

## 6. PLC Logic Summary (Ladder Logic Outline)

The PLC program is structured in TIA Portal V17 using Function Blocks (FBs) for modularity. Key program blocks include:

- FB1 — Belt Control: START/STOP logic, E-Stop interlock, VFD speed setpoint via analogue output
- FB2 — Sensor Processing: Reads all DIs, debounces with 10 ms filter timers, sets zone flags
- FB3 — Weight Inspection: Reads AI, applies limits, sets/clears Reject flag
- FB4 — Sorting Control: Monitors PS3, triggers solenoid output on Reject flag, uses TON timer for solenoid dwell time (500 ms)
- FB5 — Fault Management: Jam detection, E-Stop logic, alarm bits, fault reset sequence
- FC1 — HMI Communication: Writes counter values, alarm states, and belt speed to HMI data blocks

## 7. Expected System Performance

| Parameter                 | Value              | Notes                        |
|---------------------------|--------------------|------------------------------|
| Belt speed range          | 0.1 – 1.0 m/s      | VFD adjustable               |
| Throughput (max)          | 2,400 objects/hour | At 0.5 m/s, 200mm spacing    |
| Weighing accuracy         | ±2 g               | Load cell + averaging filter |
| Object detection response | < 5 ms             | PLC scan time + sensor delay |
| Sorting accuracy          | 99.8%              | Based on sensor specs        |
| System uptime target      | > 98.5%            | Planned maintenance 1.5%     |
| Power consumption         | ~1.2 kW            | At full load, 0.5 m/s        |

## 8. Conclusion

The automated conveyor belt system presented in this report integrates industrial-grade sensors, a Siemens S7-1200 PLC, VFD motor control, and pneumatic sorting into a coherent, high-performance automation solution. The system's modular design and use of standard industrial components ensure ease of maintenance, scalability, and reliability. The flowchart and circuit diagrams provide a complete implementation blueprint, ready for TIA Portal programming and physical commissioning.

## References

1. Siemens AG. (2024). SIMATIC S7-1200 System Manual. Retrieved from support.industry.siemens.com
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